Diagnosis, Not Architecture

UrbanTwin Sim2Real LiDAR Challenge, LUMPI and V2X-Real tracks

First place on both tracks. Neither needed a new architecture.

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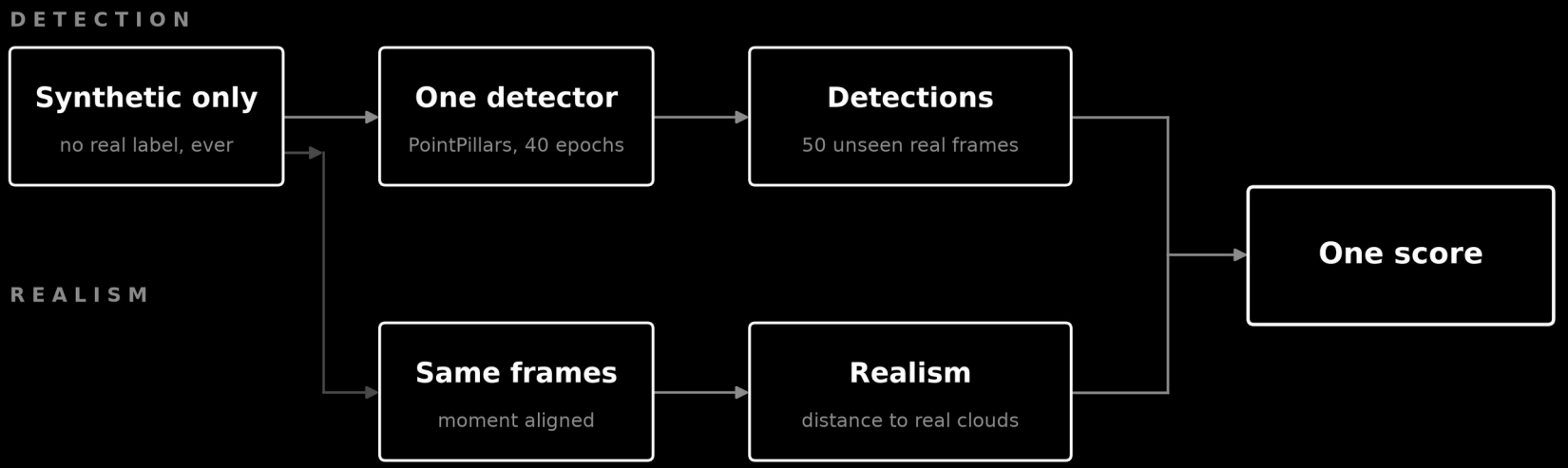
DriveX Workshop, ECCV 2026

One detector, two things scored

Train on synthetic LiDAR only. No real label ever enters the loss.

Evaluate on 50 unseen real frames, and submit 50 synthetic frames scored for distributional realism.

Detection is divided by 0.7, so an mAP of 0.7 saturates that half. Mean AP divides by three classes on V2X and seven on LUMPI.




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combined  =  0.6 × (mAP / 0.7)   +   0.4 × realism
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Same submission is scored on both halves.

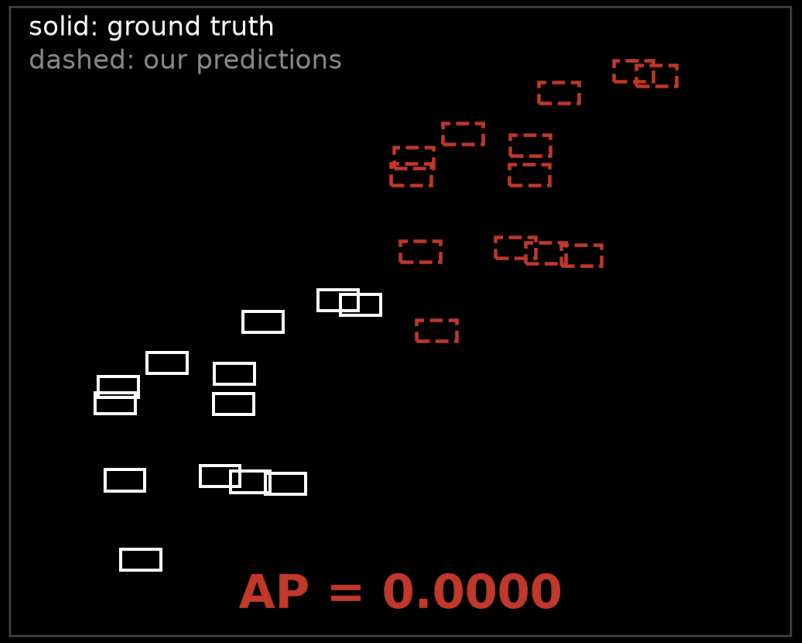
The detector was working. The coordinates were not.

Our first substantive submissions returned an AP of exactly 0.0000 for every class.

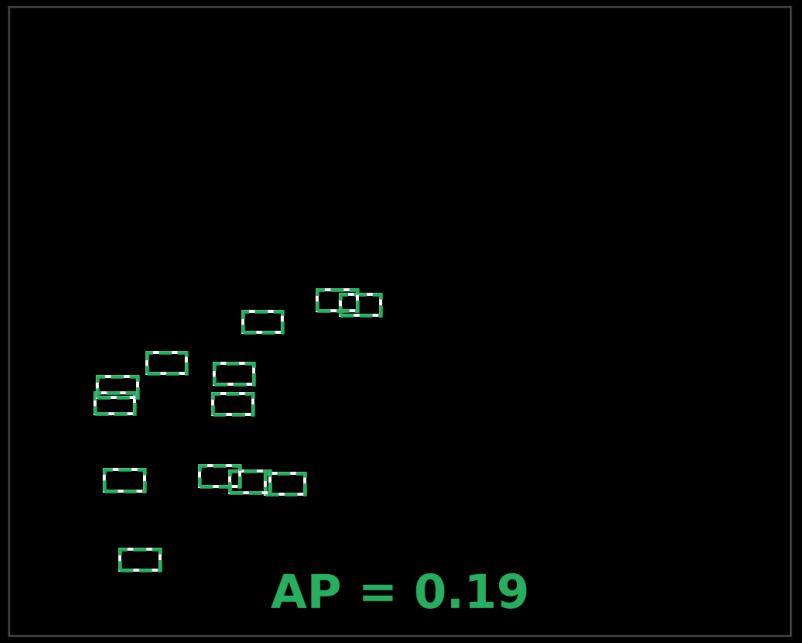
A broken detector returns 0.02, not 0.0000. Exact zeros are a format signal, not a modelling one.

Ground truth was in the dataset world frame; the released test frames are in each frame's own LiDAR frame.

Predictions in the LiDAR frame



After the inverse transform



Combined 0.3520 to 0.5147 in a single submission.

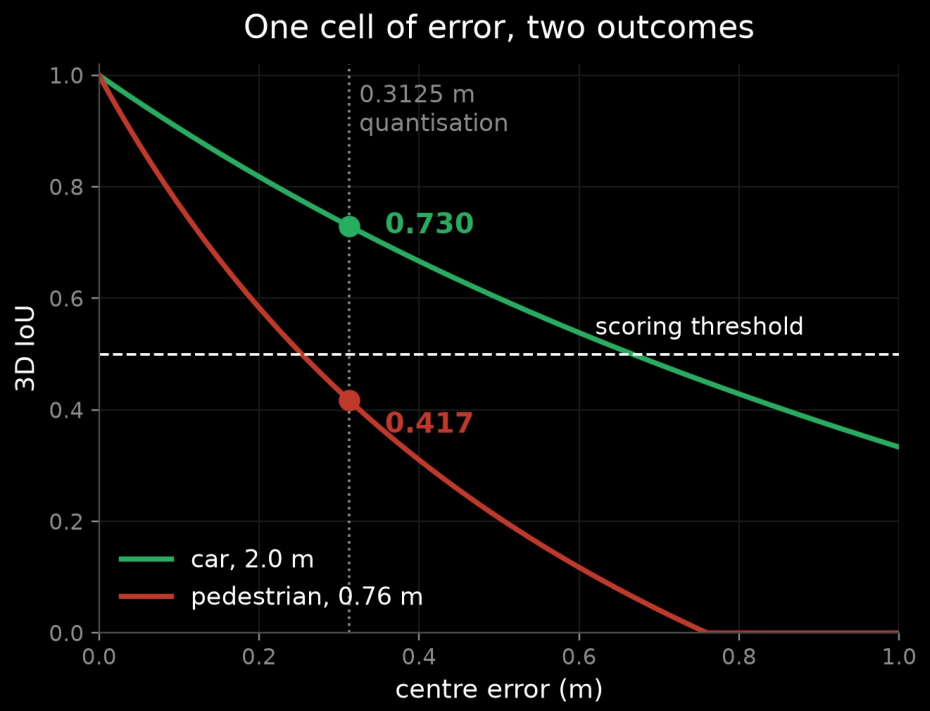
Small objects fail on geometry, not on recognition

V2X: a single global 0.3 IoU threshold suppressed real pedestrians, because people stand a metre apart.

LUMPI: Person recall was 86 percent at IoU 0.25, but median IoU against matched ground truth was 0.461.

A 0.15625 m cell with stride 2 quantises centres to 0.3125 m. A car tolerates 0.667 m of error, a pedestrian 0.253 m.





V2X: per-class suppression. LUMPI: re-centre on the points inside the box, Person AP up 86 percent.

One of the four realism metrics has a closed form

Frechet Point cloud Distance reads only the mean and the 3 x 3 covariance of the xyz coordinates.

Our clouds were compressed against the reference: 1.6 percent in x, 6.4 in y, 19.8 in z, with a 0.39 m offset in y.

A per-axis affine correction moves the metric directly. No search, no retraining, no new model.

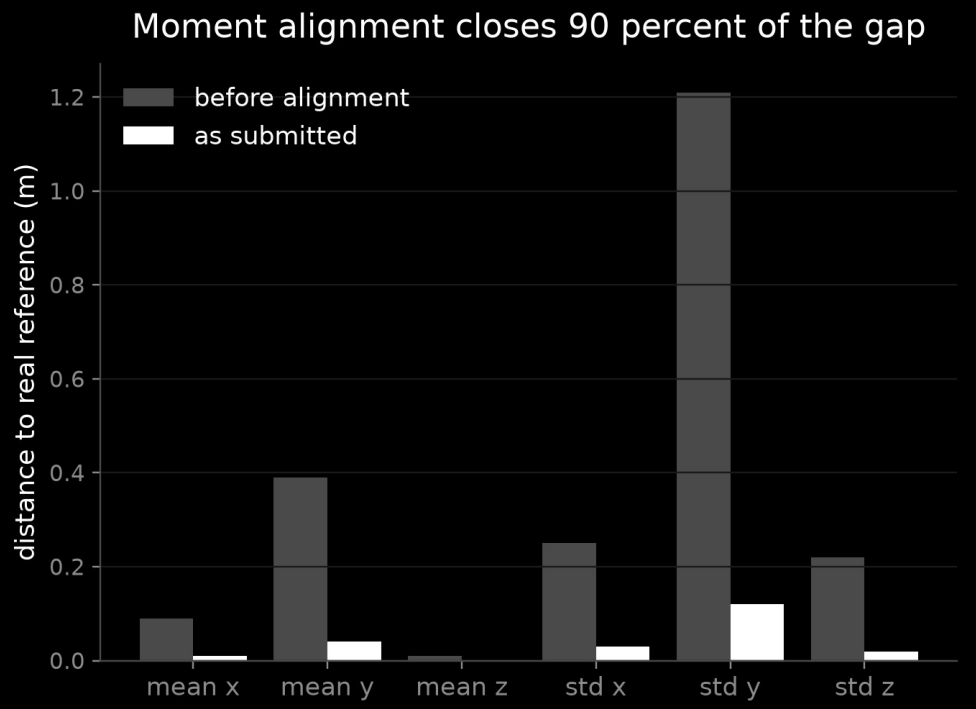
What the metric actually reads

mean
3 numbers

covariance
6 numbers

9 free parameters

so it can be solved, not searched



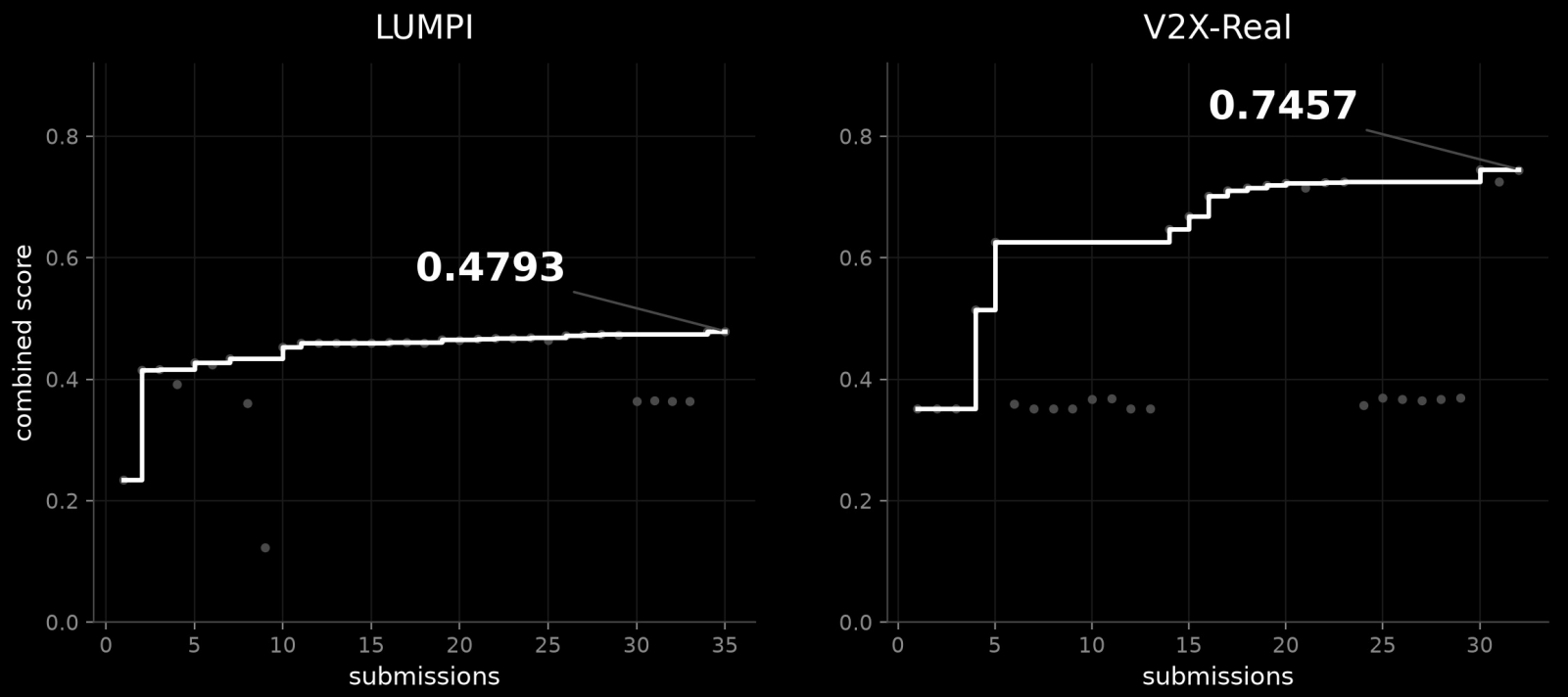
Realism 0.8800 to 0.9251. Reference moments from held-out public frames, never the hidden set.

Where the score actually came from

Every step in the best-so-far line is a diagnostic fix. None of them needed a new architecture.

Seven separate retraining runs on LUMPI came back neutral or negative.

Each stage was isolated on the evaluation server, with every untouched class acting as a control.



First on both tracks. Seven retraining runs were neutral or negative; compute was never the constraint.

Thank you

Code, weights and inference outputs are public.

github.com/amanupg/urbantwin-sim2real-lumpi

github.com/amanupg/urbantwin-sim2real-v2x

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LUMPI

V2X-Real